

CO4 : ASSESSMENT TOOL 3: Reverse Engineering Task

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1558

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects. (BL4)*

Assessment Tool Weightage: 20%

SET 1

1. A big data platform collects billions of website clicks, mobile interactions, and customer transactions every day. Reverse engineer the probable distributed storage system, ingestion framework, and batch-processing tools used. Infer how structured and semi-structured data are separated and indexed for analysis. Identify the likely stream-processing engines supporting real-time analytics and reporting. Explain the scalability and partitioning strategies probably implemented for handling massive workloads. Describe how dashboards and business intelligence systems are likely connected to the analytics layer.
2. An e-commerce platform tracks cart additions, abandoned checkouts, product views, and payment behavior across regions. Reverse engineer the likely event streaming tools, customer profiling databases, and recommendation workflow used. Infer how session data, clickstream logs, and transaction records are synchronized for real-time personalization. Identify probable caching systems, distributed query engines, and machine learning stages involved. Explain how structured order data and semi-structured browsing data are merged for analytics. Deduce the scalability and fault-tolerance decisions likely implemented. Describe how dashboards and KPI monitoring are probably generated from the pipeline.
3. A big data healthcare system processes patient histories, diagnostic images, wearable sensor streams, and prescription records. Reverse engineer the probable hybrid database architecture managing structured and unstructured healthcare information. Infer the likely data integration tools connecting hospitals, laboratories, and insurance systems. Identify the streaming and machine learning frameworks probably used for disease prediction and monitoring. Explain how privacy, encryption, and compliance requirements are likely enforced in the platform. Describe the analytics pipeline that may support clinical decision-making and population health studies.
4. A smart city platform gathers traffic sensor feeds, CCTV metadata, weather updates, and public transport activity continuously. Reverse engineer the probable edge computing setup, distributed messaging system, and centralized storage design. Infer how geospatial analytics and streaming computations are handled for congestion prediction. Identify the likely databases used for time-series and location-aware queries. Explain how batch and real-time analytics are combined for urban planning insights. Deduce the security and access-control measures likely protecting citizen and infrastructure data. Describe how visualization systems probably deliver live operational intelligence.
5. A healthcare analytics system integrates patient records, wearable device readings, lab reports, and appointment histories. Reverse engineer the probable hybrid storage

architecture handling sensitive structured and unstructured medical data. Infer the likely ETL pipelines and interoperability standards connecting hospitals and clinics. Identify how real-time monitoring and predictive health analytics are probably implemented. Explain the role of distributed processing frameworks in population-level analysis. Deduce the encryption, compliance, and audit mechanisms likely enforced for data governance. Describe how machine learning models may support diagnosis risk scoring and treatment recommendations.

1. Website Clickstream Big Data Platform

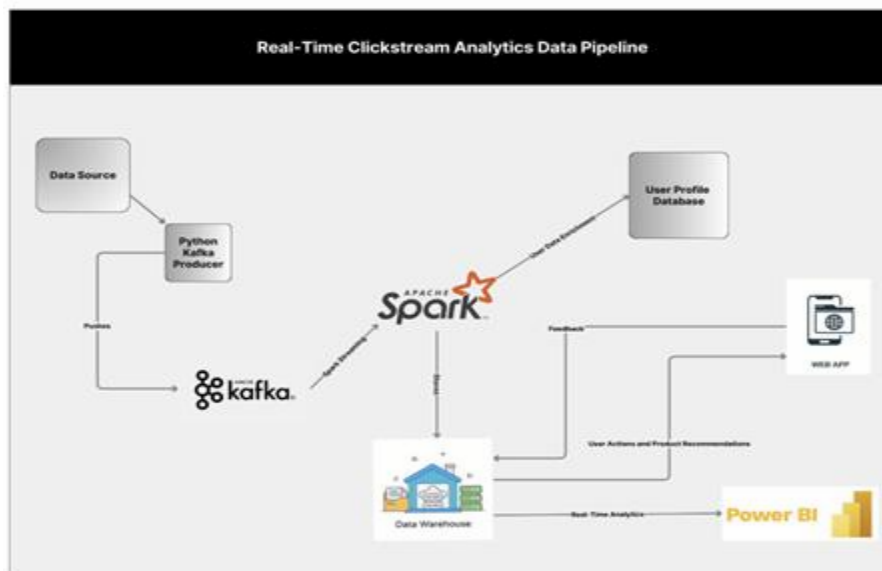
Problem Statement

A big data platform collects billions of website clicks, mobile interactions, and customer transactions every day. Reverse engineer the probable technologies used for storage, ingestion, processing, analytics, and reporting.

Objective

Infer the likely distributed architecture that supports large-scale real-time and batch analytics.

Probable Architecture



Architecture Components

Layer	Probable Technology
Data Sources	Website, Mobile App, Payment System
Ingestion	Apache Kafka
Stream Processing	Apache Flink, Spark Streaming
Batch Processing	Apache Spark
Storage	HDFS, Amazon S3
Search Index	Elasticsearch
Dashboard	Power BI, Tableau

Distributed Storage

The platform most likely separates data into different storage systems:

- Structured Data: Sales records stored in data warehouses.
- Semi-Structured Data: Clickstream logs stored in HDFS or S3.
- Search Index: Elasticsearch for fast searching.

Data Separation

Data Type	Storage
Transactions	SQL Warehouse
Click Logs	HDFS
Mobile Events	S3
Search Index	Elasticsearch

Stream Processing

Likely engines include:

- Apache Flink
 - Spark Streaming
- These process events immediately for live analytics.

Scalability Strategy

- Data partitioning by user ID and time.
- Kafka topic partitions.
- Spark distributed execution.
- Auto-scaling cloud clusters.

Dashboard Integration

Business intelligence tools connect through:

- SQL connectors
- Elasticsearch APIs
- Spark SQL

Benefits

- Real-time reporting
- Fast customer insights
- High scalability

2. E-Commerce Personalization Platform

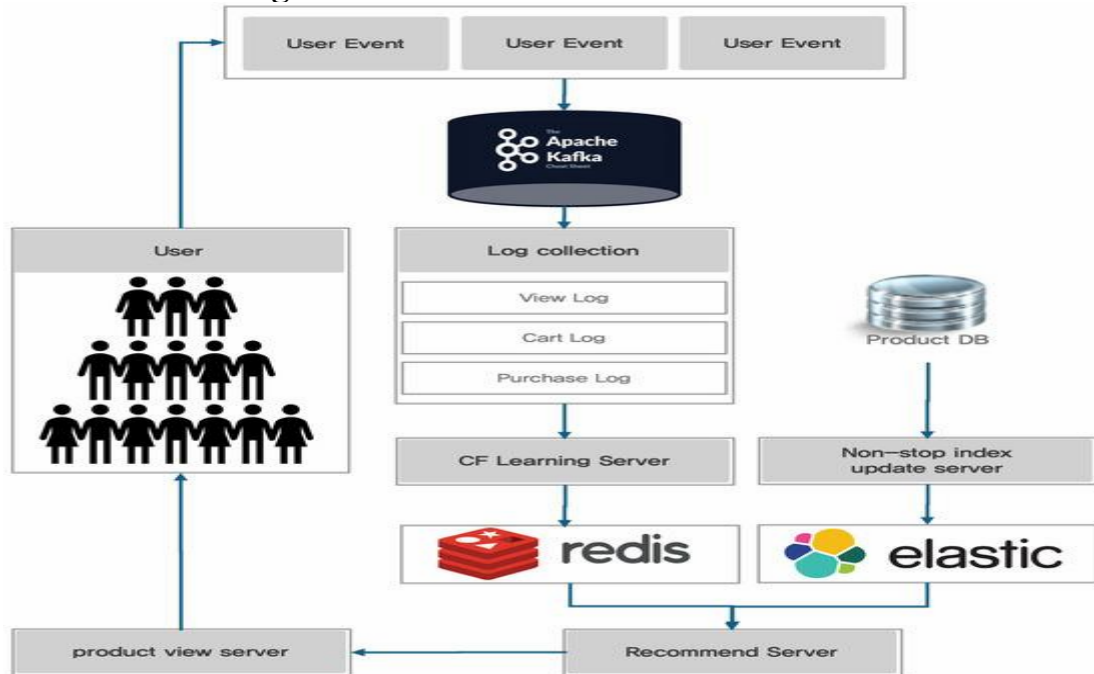
Problem Statement

An e-commerce platform tracks product views, cart additions, abandoned checkouts, and payments across regions.

Objective

Reverse engineer the event streaming, recommendation, and analytics workflow.

Architecture Diagram



Architecture Components

Component	Technology
Event Streaming	Apache Kafka
Session Cache	Redis
Storage	Data Lake
Query Engine	Trino
ML Platform	Spark MLlib

Event Streaming

Every customer action becomes an event.

Examples:

- Product View
- Add to Cart
- Checkout
- Payment

Kafka streams these events instantly.

Customer Profiling

The platform probably maintains customer profiles containing:

- Purchase history
- Browsing history
- Wishlist
- Preferences

Session Synchronization

Real-time synchronization combines:

- Clickstream logs
- Session IDs
- Orders
- Payments

Caching

Likely cache:

- Redis
- Purpose:
 - Fast recommendations
 - Active sessions
 - Frequently viewed products

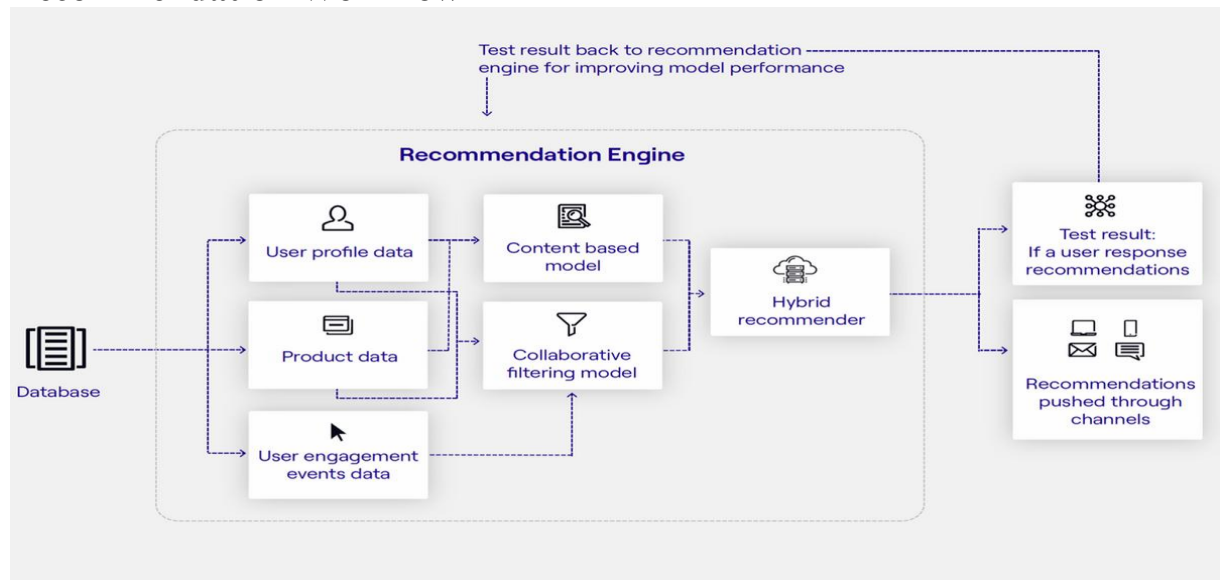
Distributed Query Engine

Probable engines:

- Trino
- Presto

These query multiple data sources simultaneously.

Recommendation Workflow



Machine Learning stages:

- Feature Engineering
- Collaborative Filtering
- Content-Based Filtering
- Hybrid Recommendation

Data Integration

Structured

Orders

Payments

Customer Profile

Semi-Structured

Click Logs

Browsing Events

Session Data

Scalability

Likely implementation:

- Kafka partitions
- Distributed Spark clusters
- Replicated Redis
- Cloud auto-scaling

KPI Dashboard

Dashboards probably display:

- Conversion Rate
- Cart Abandonment
- Revenue
- Active Users

Using:

- Power BI
- Grafana

Benefits

- Personalized shopping
- Faster recommendations
- Higher conversions

3. Healthcare Big Data Platform

Problem Statement

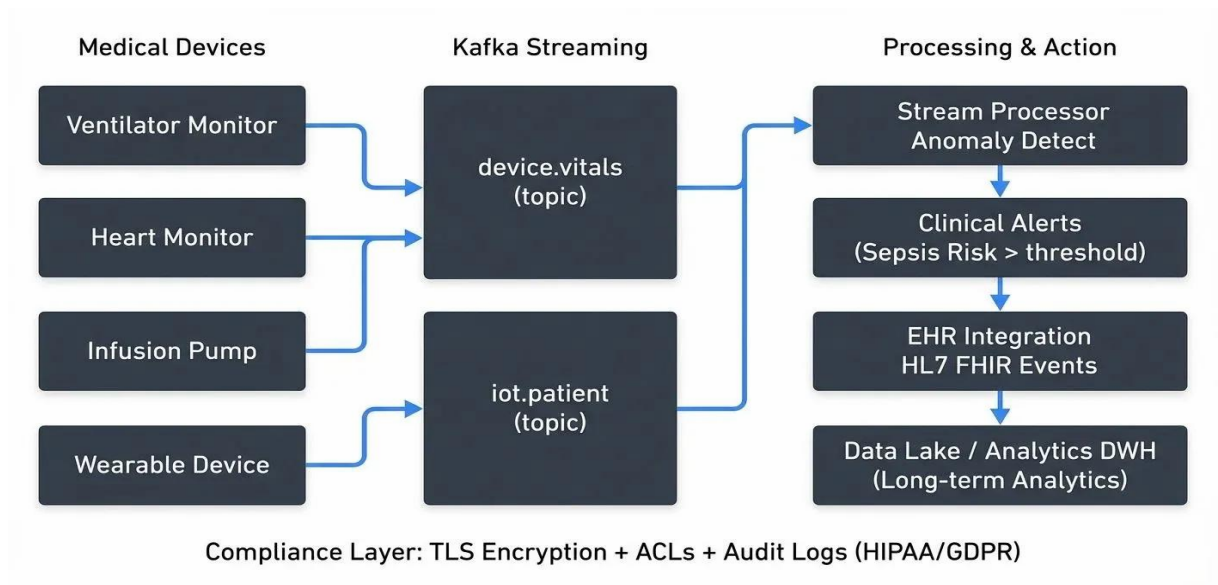
A healthcare platform processes:

- Patient histories
- Diagnostic images
- Wearable sensors
- Prescription records

Objective

Reverse engineer the hybrid storage and analytics platform.

Architecture Diagram



Hybrid Database Architecture

Data	Storage
Patient Records	PostgreSQL
Images	Object Storage
Sensor Data	Time-Series Database
Prescriptions	SQL Database

Data Integration

Hospitals likely connect using:

- HL7
 - FHIR APIs
 - Apache NiFi
- These standardize healthcare data exchange.

Streaming Framework

Wearable devices continuously send:

- Heart Rate
- Blood Pressure
- Oxygen Level

Likely tools:

- Kafka
- Spark Streaming

Machine Learning

Models probably include:

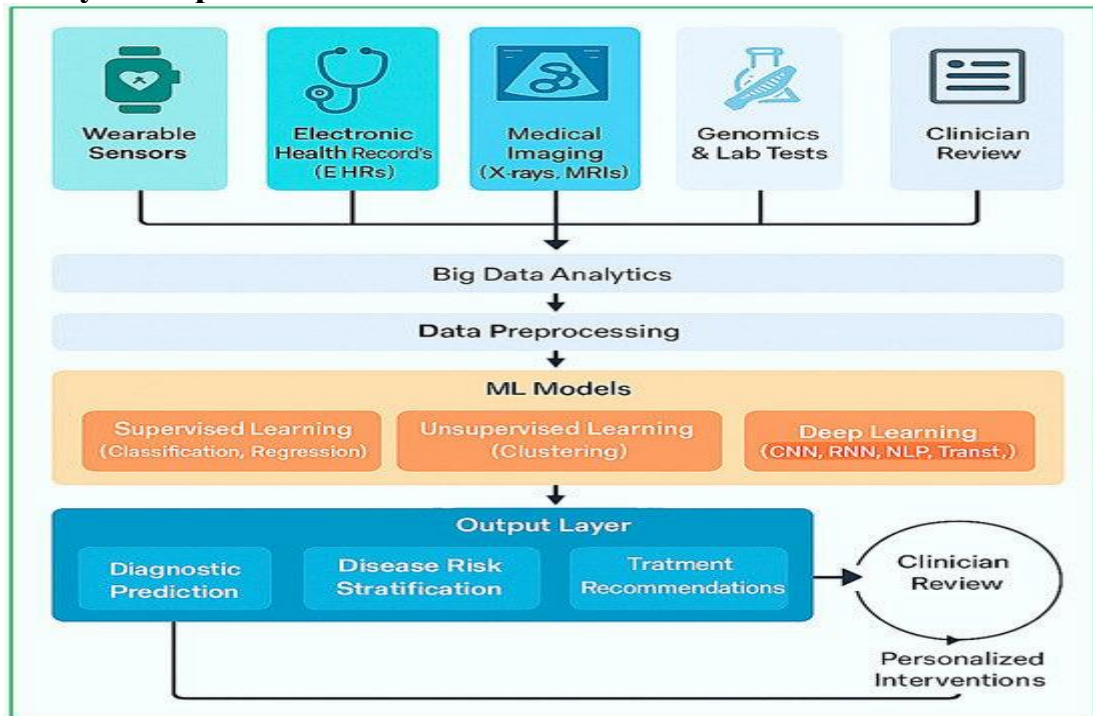
- Random Forest
- XGBoost
- CNN for medical images
- LSTM for patient monitoring

Privacy and Compliance

Security likely includes:

- AES-256 Encryption
- TLS
- Role-Based Access Control
- HIPAA compliance
- Audit logs

Analytics Pipeline



Workflow:

1. Collect patient data.
2. Clean data.
3. Store securely.
4. Run ML models.
5. Generate alerts.
6. Display clinician dashboards.

Benefits

- Better diagnosis
- Faster treatment
- Population health monitoring

4. Smart City Big Data Platform

Problem Statement

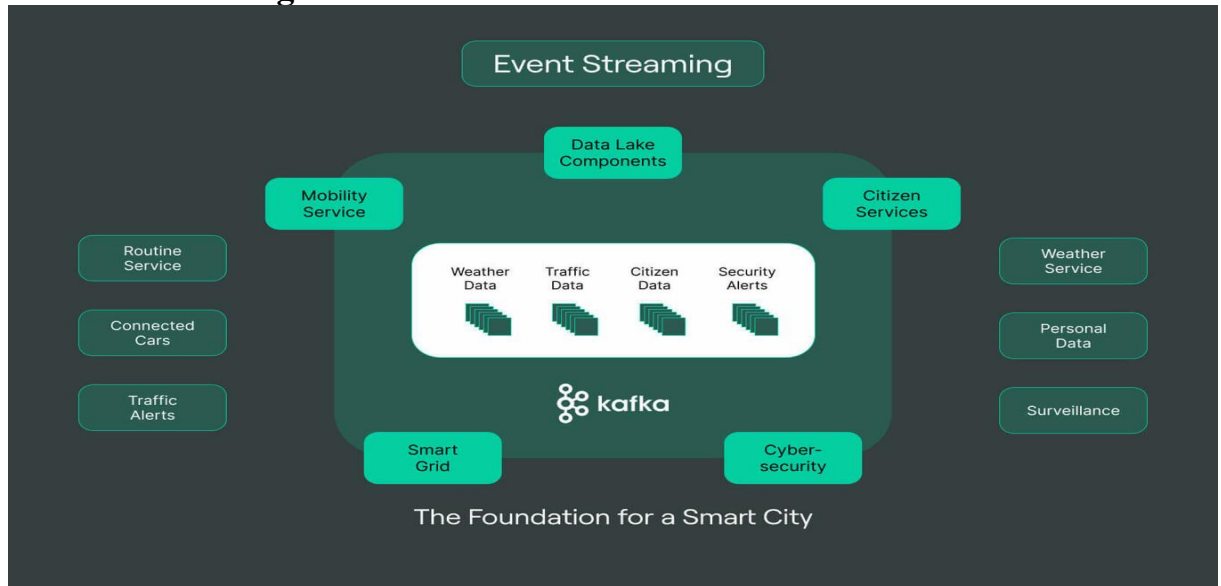
A smart city collects:

- Traffic sensors
- CCTV metadata
- Weather updates
- Public transport activity

Objective

Reverse engineer the edge computing and analytics architecture.

Architecture Diagram



Edge Computing

Edge devices probably perform:

- Video filtering
 - Traffic counting
 - Initial processing
- This reduces cloud workload.

Distributed Messaging

Likely system:

- Apache Kafka

Purpose:

- Continuous event delivery
- Reliable messaging

Central Storage

Probable storage:

- HDFS
- Cloud Data Lake

Geospatial Analytics

Likely tools:

- Apache Sedona
- PostGIS

Applications:

- Route optimization
- Traffic prediction
- Emergency response

Time-Series Database

Likely database:

- InfluxDB
- TimescaleDB
- Used for:
- Sensor readings
- Weather data

Batch and Real-Time Analytics

Type	Technology
Batch	Spark
Real-Time	Flink

Security

Likely protection includes:

- VPN
- IAM
- Encryption
- RBAC
- API Gateway

Visualization



Live dashboards probably show:

- Traffic congestion
- Bus movement
- Weather alerts
- CCTV statistics

Benefits

- Reduced congestion
- Faster emergency response
- Better urban planning

5. Healthcare Analytics System

Problem Statement

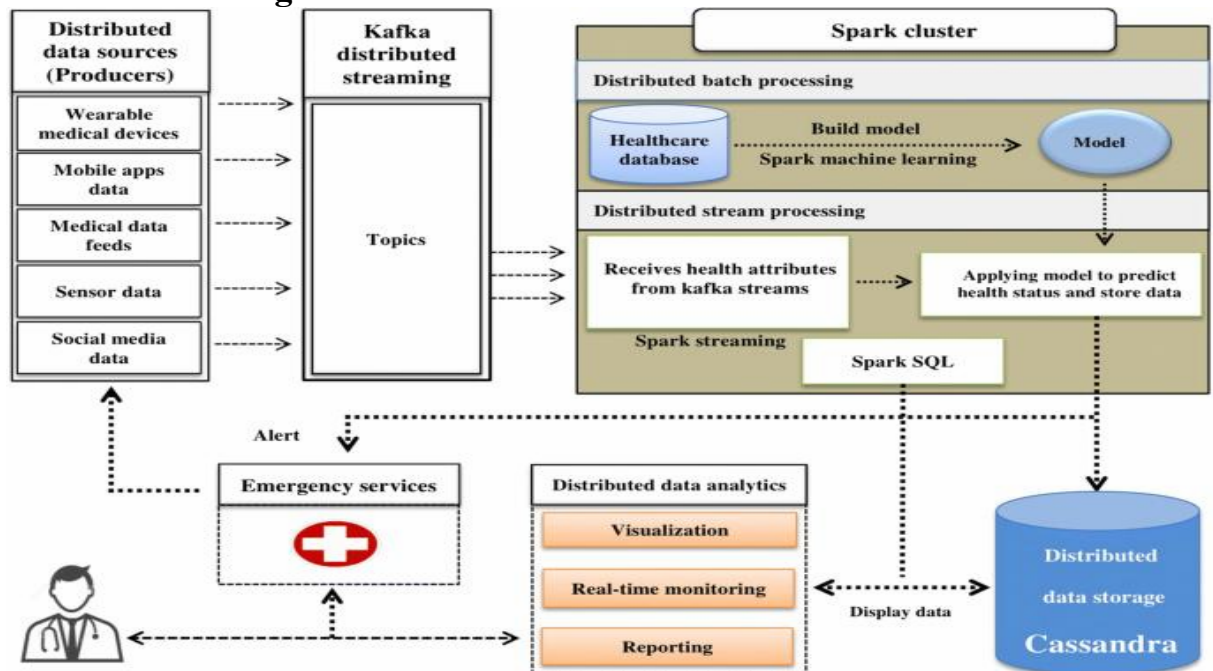
A healthcare analytics platform integrates:

- Patient records
- Wearable devices
- Lab reports
- Appointment history

Objective

Reverse engineer the analytics and governance architecture.

Architecture Diagram



Hybrid Storage

Data

Patient Records

Lab Reports

Wearables

Medical Images

Storage

PostgreSQL

Document Storage

Time-Series DB

Object Storage

ETL Pipeline

Likely ETL tools:

- Apache NiFi
- Talend

Workflow:

1. Extract hospital data.
2. Transform formats.
3. Load into central platform.

Interoperability Standards

Hospitals probably exchange data using:

- HL7
- FHIR
- DICOM

Real-Time Monitoring

Wearable devices continuously send:

- Heart Rate
- Blood Pressure
- Oxygen Level

Streaming technologies:

- Kafka
- Spark Streaming

Distributed Processing

Spark likely performs:

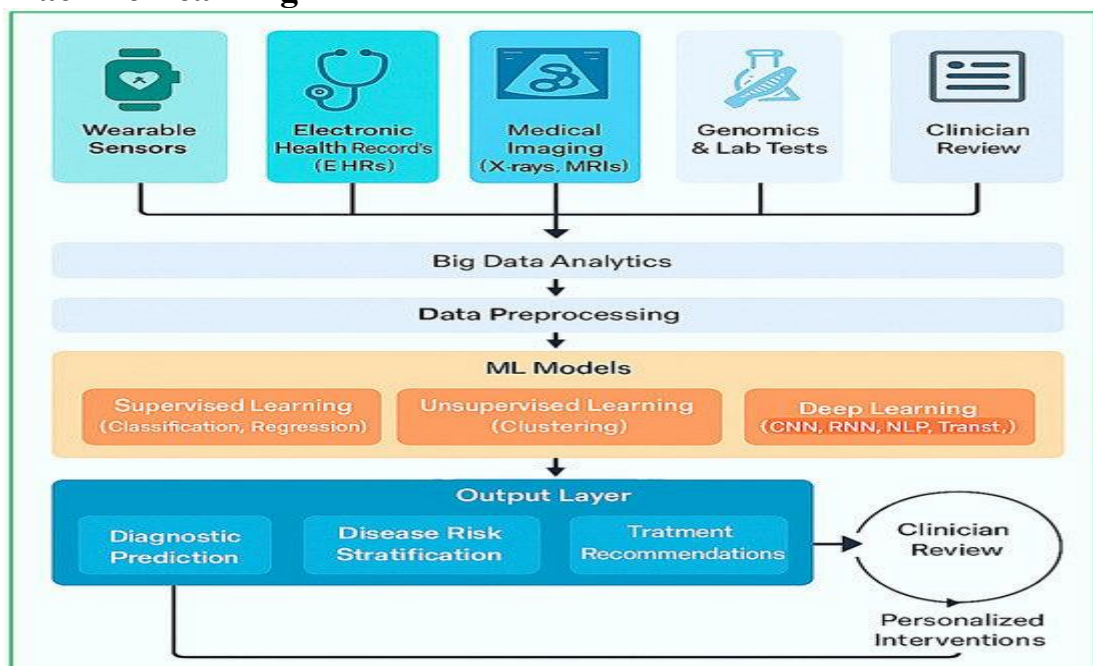
- Population analysis
- Risk prediction
- Hospital resource planning

Security

Likely governance includes:

- AES-256 Encryption
- TLS
- HIPAA Compliance
- Audit Logs
- RBAC

Machine Learning



- Models may support:
- Disease risk scoring
 - Readmission prediction
 - Treatment recommendation
 - Patient prioritization

Benefits

- Better healthcare decisions
- Improved patient monitoring
- Secure data governance

Comparison of All Reverse-Engineered Architectures

Scenario	Streaming	Storage	ML	Dashboard
Website Analytics	Kafka	HDFS	Spark ML	Power BI
E-Commerce	Kafka	Data Lake	Recommendation Engine	Grafana
Healthcare	Kafka	Hybrid DB	CNN, LSTM	Clinical Dashboard
Smart City	Kafka	HDFS	Traffic Prediction	GIS Dashboard
Healthcare Analytics	Kafka	Hybrid Storage	Risk Scoring	Power BI

Common Technologies Across All Platforms

Category	Technologies
Ingestion	Apache Kafka, Apache NiFi
Stream Processing	Apache Flink, Spark Streaming
Batch Processing	Apache Spark
Storage	HDFS, Amazon S3, PostgreSQL
Databases	Cassandra, Redis, Elasticsearch
Machine Learning	TensorFlow, Spark MLlib, Scikit-learn
Visualization	Power BI, Grafana, Tableau
Security	AES-256, TLS, RBAC, IAM

Conclusion

The reverse-engineered architectures indicate that modern big data platforms rely on a combination of distributed storage (HDFS/S3), Apache Kafka for event ingestion, Spark/Flink for real-time and batch processing, and machine learning for predictive analytics. By integrating structured and unstructured data with secure governance mechanisms, these platforms achieve high scalability, fault tolerance, low latency, and actionable insights across industries such as web analytics, e-commerce, healthcare, and smart cities.

